Chapter 16. A Taste of Topology

Compactness and connectedness of matrix groups

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A first taste of topology for matrix groups, assuming only the analysis of $\mathbb{R}^n$. Identify $\mathfrak{M}_{n,n}(\mathbb{R}) = \mathbb{R}^{n^2}$ and $\mathfrak{M}_{n,n}(\mathbb{C}) = \mathbb{C}^{n^2} = \mathbb{R}^{2n^2}$ (coordinates in any fixed order), with the Euclidean norm, distance and continuity.

16.1 Matrix group isomorphisms

Motivation. Matrix groups live in Euclidean space, so we may ask which of their properties are topological. Homeomorphisms (here informally "same-shape maps") that are also group isomorphisms identify matrix groups both as groups and as shapes.

Exercise 16.1.1

Under $\mathbb{C}^n = \mathbb{R}^{2n}$, the Hermitian dot product of $\mathbb{C}^n$ and the dot product of $\mathbb{R}^{2n}$ give the same norm (and metric).

Solution. $|a+ib|^2 = a^2 + b^2$ coordinatewise (compare Remark 10.2.20). □

Example 16.1.2

$\det : \mathfrak{M}_{n,n}(\mathbb{R}) \to \mathbb{R}$ is a polynomial map, hence continuous, so $\mathrm{GL}_n(\mathbb{R}) = \det^{-1}(\mathbb{R} - \{0\})$ is an open subset of $\mathbb{R}^{n^2}$; likewise $\mathrm{GL}_n(\mathbb{C})$ is open in $\mathbb{R}^{2n^2}$.

Example 16.1.3

$\mathrm{SL}_n(\mathbb{R}) = \det^{-1}(1)$ is a closed subset of $\mathbb{R}^{n^2}$; it is a **closed subgroup** of $\mathrm{GL}_n(\mathbb{R})$. Likewise $\mathrm{SL}_n(\mathbb{C})$ is a closed subgroup of $\mathrm{GL}_n(\mathbb{C})$.

Example 16.1.4

$\mathbf{O}(n) = \{A : A^tA - I = 0\}$ is the common zero set of n^2 polynomials in n^2 variables, hence closed; it is a closed subgroup of $\mathrm{GL}_n(\mathbb{R})$ (and a closed subset of $\mathfrak{M}_{n,n}(\mathbb{R})$).

Exercise 16.1.5

$\mathbf{U}(n)$ is a closed subgroup of $\mathrm{GL}_n(\mathbb{C})$.

Solution. $A^*A = I$ is a system of polynomial equations in the real and imaginary parts of the entries. □

Exercise 16.1.6

(a) $\mathbf{SO}(n)$ is a closed subgroup of $\mathbf{O}(n)$.
(b) $\mathbf{SU}(n)$ is a closed subgroup of $\mathbf{U}(n)$.

Solution. Each is cut out by an additional polynomial equation $\det A = 1$ (real and imaginary parts). □

Question 16.1.7

For $F = \mathbb{R}$ in Exercise 12.4.12, is $\operatorname{im}\psi$ a closed subgroup of $\mathrm{SL}_3(\mathbb{R})$?

(It is, but this needs more than these notes provide: e.g. that $\operatorname{im}\psi$ is the identity component of the group preserving a quadratic form of signature $(2,1)$.)

Definition 16.1.8 ("Same shape")

For $X \subseteq \mathbb{R}^r$, $Y \subseteq \mathbb{R}^s$, a bijection $f : X \to Y$ with f and f^{-1} continuous is a **homeomorphism** ("same-shape map"); then X and Y have the **same shape** (are homeomorphic).

Exercise 16.1.9

"Same shape" is an equivalence relation.

Solution. Identities, inverses and composites of homeomorphisms are homeomorphisms. □

Exercise 16.1.10

If $f : X \to Y$ is a homeomorphism and $Z \subseteq X$: (a) Z open in $X \Rightarrow f(Z)$ open in Y; (b) Z closed in $X \Rightarrow f(Z)$ closed in Y.

Solution. $f(Z) = (f^{-1})^{-1}(Z)$, and preimages of (relatively) open/closed sets under the continuous map f^{-1} are open/closed. □

Example 16.1.11

$(a,b) \subseteq \mathbb{R}$ and $X = \{(x,1)^t : a < x < b\} \subseteq \mathbb{R}^2$ have the same shape, although (a,b) is open in $\mathbb{R}$ and X is not open in $\mathbb{R}^2$. Any two open intervals have the same shape.

Example 16.1.12

$\exp : \mathbb{R} \to (0,\infty)$ is a homeomorphism (Example 11.5.7).

Example 16.1.13

$\tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \to \mathbb{R}$ is a homeomorphism.

Exercise 16.1.14

$[a,b)$ and $[a,\infty)$ have the same shape.

Solution. $x \mapsto a + \tan\left(\frac{\pi}{2} \cdot \frac{x-a}{b-a}\right)$ is a continuous bijection $[a,b) \to [a,\infty)$ with continuous inverse. □

So openness, closedness and boundedness are not invariants of shape.

Exercise 16.1.15

A circle and an ellipse have the same shape. What about a circle and a square?

Solution. $(x,y) \mapsto (ax,by)$ maps the unit circle onto $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The square (its boundary) also has the same shape as the circle, via radial projection $v \mapsto v/\|v\|$ from the centre. □

Exercise 16.1.16

Explain how a topologist turns a soccer ball with a needle hole inside out.

Solution. A sphere with a (small open) hole has the same shape as a closed disk; stretch the hole until the punctured sphere is a disk, flip the disk over, and shrink the hole back. Every step is a homeomorphism. □

Exercise 16.1.17

(a) Do $\mathbb{R}$ and $\mathbb{R}^2$ have the same shape?
(b) $\mathbb{R}$ and S^1?

Solution. [Supplement] (a) No: removing one point disconnects $\mathbb{R}$ but not $\mathbb{R}^2$, and a homeomorphism restricts to a homeomorphism of the complements.

(b) No: S^1 is compact and $\mathbb{R}$ is not (compactness is preserved, Proposition 16.2.1). □

Now return to $\mathfrak{M}_{n,n}(F) = \mathbb{R}^N$.

Exercise 16.1.18

For a matrix group $G \leq \mathrm{GL}_n(F)$: (a) for fixed $g \in G$, left multiplication $\lambda_g : G \to G$ is a continuous bijection, hence (its inverse is $\lambda_{g^{-1}}$) a homeomorphism; likewise right multiplication (compare Cayley's theorem); (b) inversion $x \mapsto x^{-1}$ is a homeomorphism of G; (c) for fixed g, $\mathrm{Int}(g) : G \to G$ is a homeomorphism.

Solution. Matrix multiplication is polynomial in the entries, hence continuous. Inversion is continuous by the adjugate formula $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$ and is its own inverse. $\mathrm{Int}(g) = \lambda_g \circ \rho_{g^{-1}}$. □

Observation 16.1.19

If H is a subgroup of a matrix group G and $g \in G$, then H and the coset gH have the same shape.

Proof. $\lambda_g(H) = gH$, and λ_g restricts to a homeomorphism $H \to gH$ (Exercise 16.1.18(a)). ■

Definition 16.1.20 (Matrix group isomorphism)

For matrix groups $G \leq \mathrm{GL}_n(F) \subseteq \mathbb{R}^N$ and $H \leq \mathrm{GL}_m(F') \subseteq \mathbb{R}^M$ ($F, F' \in \{\mathbb{R}, \mathbb{C}\}$, possibly different), $\varphi : G \to H$ is a **matrix group isomorphism** if it is a group isomorphism and a homeomorphism; then G and H are **isomorphic as matrix groups**, $G \approx_{\mathrm{mat}} H$.

Example 16.1.21

$\exp : (\mathbb{R}, +) \to (\mathbb{R}^+, \times)$ is a matrix group isomorphism.

Exercise 16.1.22

$\mathbf{SO}(2) \approx_{\mathrm{mat}} S^1 \approx_{\mathrm{mat}} \mathbf{U}(1)$.

Solution. $R_\theta \mapsto e^{i\theta}$, i.e. $\left(\begin{smallmatrix} x & -y \\ y & x \end{smallmatrix}\right) \mapsto x+iy$, is a group isomorphism (Exercise 11.9.20) and a restriction of a linear map with linear inverse, hence a homeomorphism; $S^1 = \mathbf{U}(1)$. □

Observation 16.1.23

If $H \leq G \leq \mathrm{GL}_n(F)$ and $g \in G$, then $H \approx_{\mathrm{mat}} gHg^{-1}$.

Proof. $\mathrm{Int}(g)$ restricts to a group isomorphism $H \to gHg^{-1}$, which is a homeomorphism by Exercise 16.1.18(c). ■

16.2 Compactness and connectedness

Motivation. Compactness and connectedness are invariants of shape. $\mathbf{O}(n)$, $\mathbf{U}(n)$ and their special subgroups are compact, $\mathrm{GL}_n(\mathbb{R})$ and $\mathbf{O}(1,1)$ are not. The spectral theorems, expressing $\mathbf{SO}(n)$ and $\mathbf{U}(n)$ as unions of conjugates of a connected torus, show that these groups are connected.

In $\mathbb{R}^N$, X is **compact** iff it is closed and bounded, and (for the sets considered here) **connected** means arcwise connected.

Proposition 16.2.1

If $f : X \to Y$ is continuous ($X \subseteq \mathbb{R}^N$, $Y \subseteq \mathbb{R}^M$): (a) X compact $\Rightarrow$ $f(X)$ compact;
(b) X connected $\Rightarrow$ $f(X)$ connected.

Proof. Well known from analysis. ■

So compactness and connectedness are invariants of shape, independent of the ambient $\mathbb{R}^N$.

Example 16.2.2

$\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{SL}_n(\mathbb{R})$ ($n \geq 2$) are not bounded in $\mathbb{R}^{n^2}$ (e.g. $\mathrm{diag}(t, t^{-1}, 1, \dots)$), hence not compact; likewise $\mathrm{GL}_n(\mathbb{C})$, $\mathrm{SL}_n(\mathbb{C})$.

Example 16.2.3

For $A \in \mathbf{O}(n) \subseteq \mathbb{R}^{n^2}$, $\|A\| = \sqrt{n}$ (its columns are unit vectors, Observation 9.3.6(5)), so $\mathbf{O}(n)$ is bounded; it is closed (Example 16.1.4), hence a compact group.

Example 16.2.4

$\mathbf{SO}(n)$, a closed subgroup of $\mathbf{O}(n)$, is compact.

Exercise 16.2.5

$\mathbf{U}(n)$ and $\mathbf{SU}(n)$ are compact.

Solution. As in Examples 16.2.3, 16.2.4: $\|A\| = \sqrt{n}$ for $A \in \mathbf{U}(n)$ (Exercise 10.6.9(5), Exercise 16.1.1), and they are closed (Exercises 16.1.5, 16.1.6). □

Example 16.2.6

(a) $\mathbf{SO}^\circ(1,1)$ is unbounded in $\mathbb{R}^4$ ($\cosh t \to \infty$), so $\mathbf{SO}^\circ(1,1)$ and $\mathbf{O}(1,1)$ are not compact. (b) Alternatively, $\mathbf{SO}^\circ(1,1) \approx_{\mathrm{mat}} (\mathbb{R}, +)$ (Exercise 13.4.5; the inverse $\left(\begin{smallmatrix} x & y \\ y & x \end{smallmatrix}\right) \mapsto \operatorname{arsinh} y$ is continuous), and $\mathbb{R}$ is not compact. (c) Since $\mathbf{SO}(2) \approx_{\mathrm{mat}} S^1$, the compact $\mathbf{SO}(2)$ and the

non-compact $\mathbf{SO}^\circ(1,1)$ are not isomorphic as matrix groups. They are not even isomorphic as abstract groups: $\mathbf{SO}(2)$ has elements of every finite order, $\mathbf{SO}^\circ(1,1) \approx \mathbb{R}$ has no nontrivial element of finite order (compare Exercise 13.4.6).

Example 16.2.7

$\det : \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^\times = \mathbb{R} - \{0\}$ is a continuous surjection and $\mathbb{R}^\times$ is not connected; so $\mathrm{GL}_n(\mathbb{R})$ is not connected.

Exercise 16.2.8

$\mathbf{O}(n)$ and $\mathbf{O}(1,1)$ are not connected.

Solution. det maps each continuously onto $\{\pm 1\}$, which is not connected. □

Proposition 16.2.9

For $X_i \subseteq \mathbb{R}^{N_i}$, identify $X_1 \times \cdots \times X_r \subseteq \mathbb{R}^{N_1+\cdots+N_r}$. (a) If all X_i are compact, so is the product.
(b) If all X_i are connected, so is the product.

Proof. Well known. ■

Definition 16.2.10 (Torus)

$\mathbf{T} \subseteq \mathbb{R}^N$ is a **torus** if it has the same shape as $S^1 \times \cdots \times S^1$. A torus is compact and connected; $S^1 \times S^1$ has the shape of a doughnut.

Exercise 16.2.11

$\mathbf{T}_{\mathbf{U}(n)}$, $\mathbf{T}_{\mathbf{SU}(n)}$ (Definition 15.2.7) and $\mathbf{T}_{\mathbf{SO}(n)}$ (Definition 15.4.4) are tori.

Solution. The isomorphisms of Exercises 15.2.8 and 15.4.5 with products of circles are coordinate maps (with continuous inverses), hence matrix group isomorphisms, in particular homeomorphisms. □

Corollary 16.2.12 (Corollary of the spectral theorems)

$\mathbf{U}(n)$, $\mathbf{SU}(n)$ and $\mathbf{SO}(n)$ are connected groups.

Proof. We treat $\mathbf{SO}(n)$; the others follow from Corollaries 15.2.9 and 15.2.11 in the same way. By Corollary 15.4.6, $\mathbf{SO}(n) = \bigcup_{U \in \mathbf{SO}(n)} U\,\mathbf{T}_{\mathbf{SO}(n)}\,U^{-1}$. Each $U\mathbf{T}_{\mathbf{SO}(n)}U^{-1} \approx_{\mathrm{mat}} \mathbf{T}_{\mathbf{SO}(n)}$ (Observation 16.1.23) is connected (Exercise 16.2.11), and all of them contain I. A union of connected sets with a common point is connected. ■

Since $\mathbf{O}(n)$ is not connected, we look at its **connected components** (maximal connected subsets); every $X \subseteq \mathbb{R}^N$ is uniquely the disjoint union of its components (**component decomposition**).

Observation 16.2.13

(a) $\mathbf{SO}(n)$ is a connected component of $\mathbf{O}(n)$.
(b) The coset decomposition $\mathbf{O}(n) = \mathbf{SO}(n) \amalg (\mathbf{O}(n) - \mathbf{SO}(n))$ is the component decomposition of $\mathbf{O}(n)$.

Proof. $\mathbf{SO}(n)$ is connected (Corollary 16.2.12), and so is $\mathbf{O}(n) - \mathbf{SO}(n) = S \cdot \mathbf{SO}(n)$ (Observation 16.1.19). No connected subset can meet both, since det would map it continuously onto $\{\pm 1\}$. ■

Observation 16.2.14

(a) $\mathbf{SO}^\circ(1,1)$ is a connected component of $\mathbf{O}(1,1)$.
(b) The coset decomposition of $\mathbf{O}(1,1)$ modulo $\mathbf{SO}^\circ(1,1)$ (§13.4) is its component decomposition.

Proof [Supplement]. $\mathbf{SO}^\circ(1,1) \approx \mathbb{R}$ is connected, and so are its cosets (Observation 16.1.19). The four cosets are separated by the continuous functions det and $\operatorname{sign}(A_{11})$ ($|A_{11}| \geq 1$ never vanishes), which are constant on connected sets and take the four combinations of values ± 1 on the four cosets. ■

That the component containing the identity is a subgroup is no accident (proof postponed). Thus $\mathbf{O}(2)$ has the shape of two disjoint circles and $\mathbf{O}(1,1)$ that of four disjoint lines.

16.3 Selected exercises

Motivation. Three further results on connectedness: the identity component is always a normal subgroup, $\mathrm{GL}_n(\mathbb{C})$ is connected, and $\mathrm{GL}_n(\mathbb{R})$ has exactly two components, distinguished by the sign of det.

Exercise 16.3.1 (The identity component is a normal subgroup)

Let $G \leq \mathrm{GL}_n(F)$ be a matrix group and let G° be the set of $g \in G$ that can be joined to I by a continuous path in G. Show that G° is a normal subgroup of G, and that the connected components of G are the cosets xG°.

Solution [Supplement].

Step 1 (subgroup). $I \in G^\circ$ (constant path). Let $g, h \in G^\circ$ with paths $\alpha, \beta : [0,1] \to G$ from I to g and to h. By Exercise 16.1.18, multiplication and inversion are continuous, so $\gamma(t) = \alpha(t)\beta(t)^{-1}$ is a path in G from I to gh^{-1}. Hence $gh^{-1} \in G^\circ$.

Step 2 (normal). For $x \in G$, $t \mapsto x\alpha(t)x^{-1}$ is a path in G from I to xgx^{-1}, so $xG^\circ x^{-1} \subseteq G^\circ$.

Step 3 (components). G° is the (arcwise) component of G containing I. Since λ_x is a homeomorphism of G (Exercise 16.1.18), it maps components onto components, so the component of x is $\lambda_x(G^\circ) = xG^\circ$. This explains the remark after Observation 16.2.14: there G° is $\mathbf{SO}(n)$, respectively $\mathbf{SO}^\circ(1,1)$. □

Exercise 16.3.2 ($\mathrm{GL}_n(\mathbb{C})$ is connected)

Show that any $A, B \in \mathrm{GL}_n(\mathbb{C})$ can be joined by a continuous path in $\mathrm{GL}_n(\mathbb{C})$.

Solution [Supplement].

Step 1. $p(z) = \det\big((1-z)A + zB\big)$ is a polynomial in $z \in \mathbb{C}$ with $p(0) = \det A \neq 0$ and $p(1) = \det B \neq 0$. So $p \not\equiv 0$ and its zero set Z is finite, with $0, 1 \notin Z$.

Step 2. The segment $[0, w]$ meets Z only if w lies on one of the finitely many rays from 0 through points of Z; likewise for $[w, 1]$ and rays from 1. Choose $w \in \mathbb{C}$ off all these rays (and $w \notin Z$), and let γ be the polygonal path $0 \to w \to 1$; it avoids Z.

Step 3. Then $M(t) = (1-\gamma(t))A + \gamma(t)B$ is continuous, $\det M(t) = p(\gamma(t)) \neq 0$, $M(0) = A$ and $M(1) = B$.

In contrast $\mathrm{GL}_n(\mathbb{R})$ is not connected (Example 16.2.7): over $\mathbb{R}$ one cannot go around the zeros of p. □

Exercise 16.3.3 (Components of $\mathrm{GL}_n(\mathbb{R})$)

Let $\mathrm{GL}_n^{\pm}(\mathbb{R}) = \{A \in \mathrm{GL}_n(\mathbb{R}) : \pm\det A > 0\}$. Show that $\mathrm{GL}_n^{+}(\mathbb{R})$ and $\mathrm{GL}_n^{-}(\mathbb{R})$ are the connected components of $\mathrm{GL}_n(\mathbb{R})$.

Solution [Supplement].

Step 1 (QR decomposition). Let $A \in \mathrm{GL}_n^{+}(\mathbb{R})$ with columns $a_1, \dots, a_n$. Gram–Schmidt applied to $a_1, \dots, a_n$ gives an orthonormal basis $q_1, \dots, q_n$ with $a_j = \sum_{i \le j} r_{ij} q_i$ and $r_{jj} > 0$. So $A = QR$ with $Q \in \mathbf{O}(n)$ and R upper triangular with positive diagonal. As $\det R > 0$, $\det Q = \det A / \det R > 0$, so $Q \in \mathbf{SO}(n)$.

Step 2 (a path to A). By Corollary 16.2.12 there is a path σ in $\mathbf{SO}(n)$ from I to Q. Let $R_t = (1-t)I + tR$, upper triangular with diagonal entries $(1-t) + t r_{jj} > 0$. Then $t \mapsto \sigma(t) R_t$ is a path from I to A with $\det\big(\sigma(t)R_t\big) = \det R_t > 0$, so it stays in $\mathrm{GL}_n^{+}(\mathbb{R})$. Hence $\mathrm{GL}_n^{+}(\mathbb{R})$ is connected.

Step 3. With $S = \mathrm{diag}(-1, 1, \dots, 1)$, $\mathrm{GL}_n^{-}(\mathbb{R}) = S \cdot \mathrm{GL}_n^{+}(\mathbb{R})$, which is connected by Observation 16.1.19.

Step 4. A connected subset of $\mathrm{GL}_n(\mathbb{R})$ cannot meet both sets, since the continuous function det would map it onto a connected subset of $\mathbb{R}^\times$ containing a positive and a negative number. So these two sets are the components, and $\mathrm{GL}_n(\mathbb{R})^\circ = \mathrm{GL}_n^{+}(\mathbb{R})$ in the notation of Exercise 16.3.1. □